

nature of the data) is checksummed using a user-selectable algorithm. Traditional file systems that do provide checksumming have performed it on a per-block basis, out of necessity due to the volume management layer and traditional file system design. The traditional design means that certain failure modes, such as writing a complete block to an incorrect location, can result in properly checksummed data that is actually incorrect. ZFS

checksums are stored in a way such that these failure modes are detected and can be recovered from gracefully. All checksumming and data recovery is done at the file system layer, and is transparent to applications. In addition, ZFS provides for self-healing data. ZFS supports storage pools with varying levels of data redundancy, including mirroring and a variation on RAID-5. When a bad data block is detected, ZFS fetches the correct data from another redundant copy, and repairs the bad data, replacing it with the good copy.

Quick Tip

Check out a video on how ZFS Self-Healing recovers data from a hard disk which has been smashed by a hammer at <http://www.thinkdigit.com/opensolaris>

4.5 Unparalleled scalability

ZFS has been designed from the ground up to be the most scalable file system, ever. The file system itself is 128-bit, allowing for 256 quadrillion zettabytes of storage. All metadata is allocated dynamically, so no need exists to pre-allocate inodes (inode is a structure which provide details about the file such as name and attributes) or otherwise limit the scalability of the file system when it is first created. All the algorithms have been written with scalability in mind. Directories can have up to 248 (256 trillion) entries, and no limit exists on the number of file systems or number of files that can be contained within a file system.

4.6 ZFS snapshots

A snapshot is a read-only copy of a file system or volume. Snapshots can be created quickly and easily. Initially, snapshots consume no additional space within the pool. As data within the active dataset changes, the snapshot consumes space by continuing to reference the old data. As a result, the snapshot prevents the data from being freed back to the pool.

ZFS snapshots include the following features:

- Persist across system reboots.
- The theoretical maximum number of snapshots is 264.
- Use no separate backing store. Snapshots consume disk space directly